Kristian Knakkergaard Nielsen

List of Publications

Condensed Matter Theory
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Bibliographic overview (as of 12.08.2025)

- Peer-reviewed publications: **20**
  including high-impact journals: 1 Nature Physics and 3 Physical Review Letters
- First/corresponding authorships: **11**/**12**
- Single/last authorships: **6**/**2**
- Number of citations/H-index: **390**/**9**

Research interests

- Quantum simulation: ultracold atoms in optical lattices, Rydberg arrays.
- Numerical methods for quantum many-body systems: tensor networks, exact diagonalisation, Monte-Carlo simulations.
- Quantum many-body systems in and out of equilibrium: Transport, localisation, Hilbert space fragmentation, Feynman diagrams.
- Quantum optics of many-body systems: Atomic arrays, polaritons.

Education

2017–2021 **PhD in Physics**, *Aarhus University*, Aarhus, Denmark
21 January 2021

2015–2017 **Cand. scient. in Physics**, *Aarhus University*, Aarhus, Denmark
Average grade: 11.9

2012–2015 **BSc in Physics and Astronomy**, *Aarhus University*, Aarhus, Denmark
Average grade: 11.2

2009–2012 **High school**, *Thisted Gymnasium og HF-kursus*, Thisted, Denmark
Average grade: 11.9

Postdoc positions

2025– Condensed Matter Theory/Theoretical Quantum Optics/Niels Bohr International Academy, Niels Bohr Institute, University of Copenhagen, Denmark

2022–2025 Theory Division, Max Planck Institute of Quantum Optics. Garching, Germany

2021–2022 Center for Complex Quantum Systems, Aarhus University. Aarhus, Denmark

PhD

title *Dynamics of Impurities in a Bose-Einstein condensate*

supervisors Professors Thomas Pohl and Georg M. Bruun

Master's thesis

title *Topological superfluids in Fermi-Bose mixtures*

supervisor Professor Georg M. Bruun

Grants & Awards

2024 **Carlsberg Foundation Reintegration Fellowship**
The Carlsberg Foundation, Denmark

2021 **Carlsberg Foundation International Fellowship**
The Carlsberg Foundation, Denmark

2019 **Elite Research travel grant**
Ministry of Higher Education and Science, Denmark

Teaching assistant

2023 Quantum Mechanics 2. Master's course. Technical University of Munich, Germany

2015, 2016, 2018 Quantum Mechanics. Bachelor course. Aarhus University, Denmark

2018 Electromagnetism, waves and optics. Bachelor course. Aarhus University, Denmark

2014 Calculus I and II. Bachelor course. Aarhus University, Denmark

Supervision

2025 I supervised bachelor student Mattias Evans during his internship at Niels Bohr Institute, Denmark

2021–2024 I co-supervised PhD student Jens H. Nyhegn at Aarhus University, Denmark. Main supervisor: Professor Georg M. Bruun

2015–2018 I supervised projects related to the Bachelor course on Quantum Mechanics at Aarhus University

Recent & upcoming talks

Oct 2025 **Invited talk**. *Etic Tlahuilli, 2nd polariton meeting.* Mexico City, Mexico

Sep 2025 **Invited talk**. *Nordic Conference on Recent Trends in Condensed Matter Physics.* Copenhagen, Denmark

Aug 2024 **Contributed talk**. *Collective Phenomena Workshop.* Konstanz, Germany

Aug 2024 **Contributed talk**. *APS March Meeting 2024.* Minneapolis, Minnesota, USA

Aug 2023 **Invited talk**. *Impurity Physics with Cold Atoms and Ions.* Aarhus, Denmark

Reviewing activity

+10 papers for Physical Review Letters, Physical Review A, Physical Review B and SciPost Physics in the past two years

Outreach events

Apr 2021 Gave a talk at the Festival of Research (Forskningens døgn).

Academic Organisation

2025– Organized weekly Condensed Matter Theory group meetings

2023–2024 Co-organized weekly Quantum Many-Body Seminar series at Max Planck Institute for Quantum Optics

2024 Co-organized election of MPQ representative in the Scientific Council of the Max Planck Society

2021 Co-organizer of Young Atomic Opticians conference 2021. Aarhus University, Denmark

2017	Co-founder of ISR. Student organisation at Aarhus University, Denmark
2014–2017	Member of the Friday lecture committee at Aarhus University, Denmark